

BENCHMARK: INACCURATE

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2456156.6539 +/- 0.0027 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.16 +/- 0.047                  |
| Transit depth (Rp/Rs)^2                           | 2.55 +/- 1.49 [%]               |
| Orbital Inclination (inc)                         | 89.6 +/- 2.9                    |
| Airmass coefficient 1 (a1)                        | 0.147 +/- 0.072                 |
| Airmass coefficient 2 (a2)                        | 1.7 +/- 1.09                    |
| Scatter in the residuals of the lightcurve fit is | 43.48 %                         |
| Best Comparison Star                              | #1 - [551, 426]                 |
| Optimal Aperture                                  | 1.82                            |
| Optimal Annulus                                   | 6.75                            |
| Transit Duration (day)                            | 0.0423 +/- 0.0061               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.16 $\pm$ 0.047 vs 0.1326 (deviation 0.58 $\sigma$ ) |
| Duration fit vs published            | 0.0423 d vs 0.0737 d (deviation 5.16 $\sigma$ )       |
| Mid-time O-C                         | 4.6 min                                               |
| Depth SNR                            | 3.4                                                   |
| Residual scatter                     | 43.48 %                                               |

Light curve

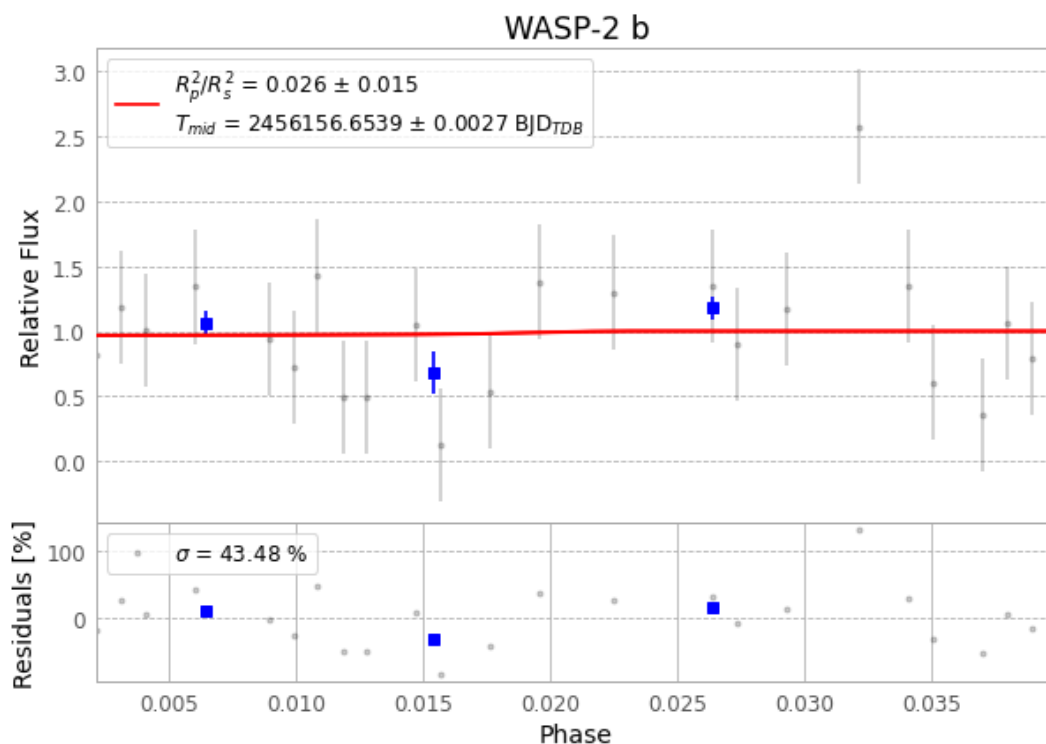


Figure 1 — FinalLightCurve\_WASP-2 b\_2012-08-16.png (SHA-256 c7033997e212e4f5...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [582, 230], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 31a31ccd35989d94...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1dc3d22

## Data provenance

44 FITS frames (29 MB), WASP-2120817032712.FITS → WASP-2120817053613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-120817.log (a6633ab97735...), FinalLightCurve\_WASP-2 b\_2012-08-16.png (c7033997e212...), FinalLightCurve\_WASP-2 b\_2012-08-16.csv (150f6e708a2a...)

## Compute resources

Wall time 87.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:13Z.